

SHRI G.S. INSTITUTE OF TECHNOLOGY AND SCIENCE, INDORE - 3

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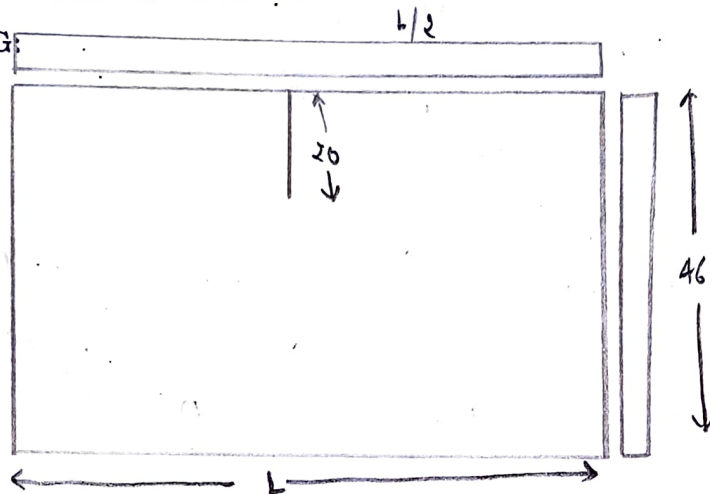
FITTING - SHOP

Roll No. : 16061

TITLE : To prepare job as per drawing.

OBJECTIVES : To familiarise with different marking, measuring and cutting tools.
: To learn different operation performed in the fitting shop.

DRAWING:



Rectangular piece with saw cut.

Raw steel : Mild steel flat

Raw material size : 60 x 48 x 6

All L's are 90°

L = free length
All dia in m.m.

Machine / Hand Tool	Particular	Specification
Marking Tool / Measuring Tools	Hand file Hock saw Bench vice Try-square Surface gauge Flat scriber Odd leg calliper Centre punch Hammer	Bastard, double cut (1.1) Fixed type solid blade Parallel jaw bench-wise

Sequence of operation:

- 1). Surface finish all edges are made $\perp$ to each other and checked whether they are true with the help of try-square. Now marking are done.
- 2). Permanent marking done with centre punch and hammer cut is made.

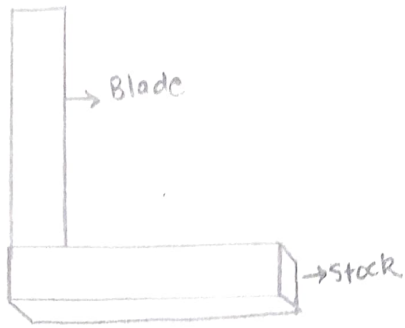
Precautions:

- 1). File should be moved across the work and be kept horizontal
- 2). Oily and dirty surface should be cleaned before filing
- 3). Job should be hold horizontal in vice and should be level with horizontal elbow.

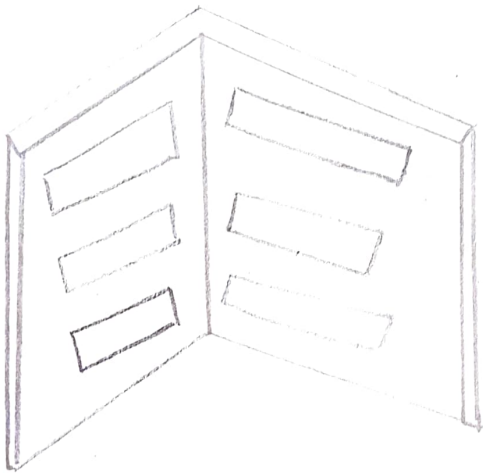
QUESTIONARIES:

Fitting 3

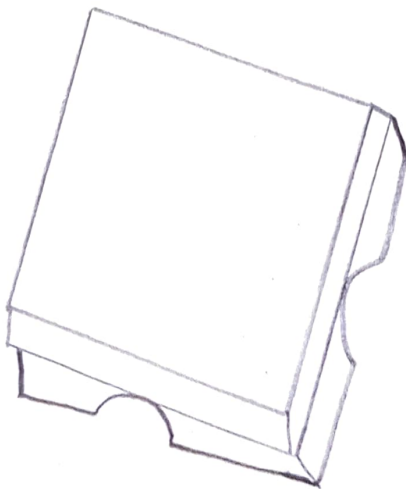
- 01 What are the different operation performed in the fitting shop ?
- 02 How the files are classified ?
- 03 What are the different part of the File ?
- 04 Give the specification of Hand Hacksaw blade ?
- 05 What are the different type of the vice and name the parts of bench vice specify the material also.
- 06 What are the uses of different files ?
- 07 Why should dirty and oily surface are cleaned before filing ?
- 08 What will happens if a file of smaller pitch is used than recommended.
- 09 State the reason why a hack-saw blade break ?
- 10 Write the use of following:
 - (i) Marking block
 - (ii) Angle plate
 - (iii) Surface plate
 - (iv) Centre punch and Dot punch
- 11 How tape are use for internal thread cutting ?



TRY-SQUARE



ANGLE PLATE



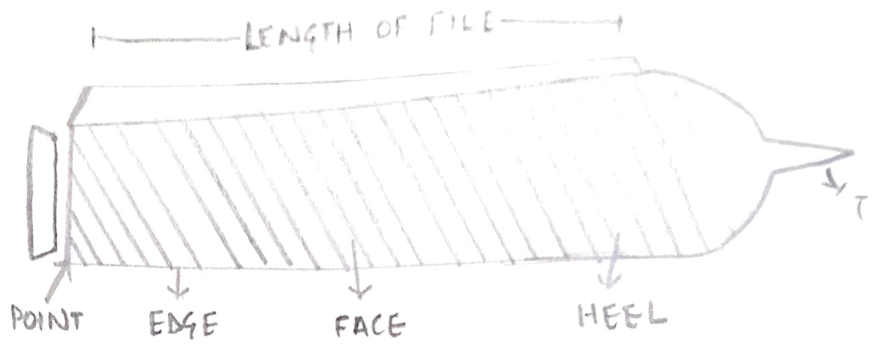
SURFACE PLATE



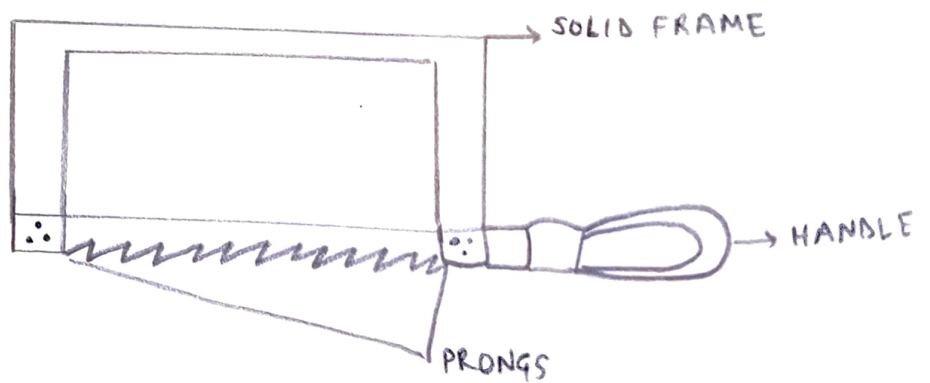
Rectangular Prism



Rectangular Prism



DIFFERENT PARTS OF A FILE



HACK SAW

Ans1: operation performed in fitting are:

- (i) chipping (ii) Filing (iii) Scrapping (iv) Grinding
(v) Sawing (vi) Marking (vii) Drilling (viii) Tapping
(ix) Reaming (x) Polishing (xi) Boring

Ans2: Files are classified as:

(i) Size: It is the distance from point to the heel without the tang

(ii) Cut of teeth:

(a) Single cut: Teeth are cut 11 to other across the file at 60° to centre line of the file termed 'flats' used on very hard metal.

(b) Double cut: 2 sets of teeth over cut teeth are cut at 60° and upper cut at $75^\circ - 80^\circ$ to the centre line.

Above two are further divide acc. to coarsness

(space b/w rows of teeth) in decreasing order of roughness they are:

- (i) Rough (R) (ii) Bastard (B) (iii) Second cut (Sc) (iv) Smooth (S)
(v) Dead smooth (DS) (vi) Super smooth (SS)

(iii) Shapes: General outline and cross-section

- (a) flat (b) hand (c) Square (d) pillar (e) Round
(f) Triangular (g) Half Round (h) Knife edge

Ans3: The parts of file are:

(i) long-pointed part which fits into handle

(ii) point-end opposite to tang

(iii) Heel next to the handle

(iv) Sage edge : side of file which has no teeth

(v) Face : The side that contains cuts and teeth

Ans 4 : specification of hand hack saw blade is

(a) length : 250 & 350 mm

(b) width : 13 & 16 mm

(c) Thickness : 0.63 & 0.8 mm

(d) Pitch : coarse teeth saw pitch = 1.8 mm

medium " " " = 1.4 mm

fine " " " = 1 mm

Ans 5 : Different types of vices are

(i) Bench (ii) leg (iii) pipe (iv) Hand (v) pen (vi) Tool maker

Bench vice has C.I body, a fixed jaw, a movable jaw both of cast steel, handle, a square, threaded screw, nut, all of mild steel, jaw plates - cast steel, protective groups / clamp made of lead fibre, tin plated

Ans 6 : (i) Flat file : Used for general work

(ii) Hand file : finishing flat surface

(iii) Square file : Filing square corners, enlarging or rectangular opening as splines and keyways.

(iv) Pillar file : Narrow working : key ways slots, grooves etc.

- (vi) Round file: Filing curved surface, enlarging round holes and forming fillets.
- (vii) Triangular file: Rectangular cuts and filing corners less than 90°
- (viii) half round file: Round cuts and filing curved surface
- (viii) Rough file: For soft metal & removing rough edges
- (ix) Knife edge file: narrow slots
- (x) Bastard file: General shaping work
- (xi) second cut file: harder metal good finish

Ans 7: If dirt and oil is present on the surface to be filed friction will $\downarrow$ and file will slip on surface resulting in non-uniform finish and irregular filing

Ans 8: If smaller pitch used filing takes much more time and effort and it is difficult to handle

Ans 9: Hacksaw blade breaks when

- (a) wrong choice of blade is made
- (b) inconveniently placed work
- (c) Unskilled operator

Ans 10: (i) Marking block: Used to mark surface and it is called precision tool

(ii) Angle plate: Used in conjunction with the surface plate for supporting work in I° position

(iii) Surface plates: Used for testing the flatness of work and marking out work

(iv) Dot punch: Used to make small punch marks on layout lines in order to make them last long

(v) Centre punch: Used to make prick punch marks larger at the centre of holes that are to be drilled